

2025

Civil Engineering Portofolio

Rizki Haikal Pradana



Summary

Hello! I'm Rizki Haikal Pradana.

I am a third-year Civil Engineering student at Universitas Gadjah Mada (UGM) with a passion for shaping the built environment through innovative and sustainable solutions.

Education



Universitas Gadjah Mada
Undergraduate Civil Engineering
2023-2027 (Expected)
Current GPA : 3.96/4.00



Pradita Dirgantara
High School (Natural Science)
2020-2023
Score : 93.49/100



Organizational Experiences



Pelayanan Komputer Sipil dan Lingkungan (PKTSL) FT UGM

Secretary General
(2025-Now)

- Manage PKTSL division activities and training, which includes Civil Engineering Drawing Tutorial with AutoCAD, and Structural Engineering Softwares with more than 100 participants.
- Upgrading PKTSL skill in computer with regular Internal House Training
- Manage national Building Information Modelling (BIM) Workshop and seminar in collaboration with BIM WIKA.

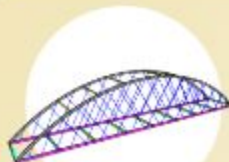


Gadjah Mada Building And Bridge

Staff of Bridge Division
(2024-Now)

- Actively involved in research, development, and innovation activities of sustainable bridge design and construction, which focuses on engineering solutions that integrate structural efficiency, environmental responsibility, and innovation.
- Represent Universitas Gadjah Mada in various national and international bridge competitions.

Interest



Structural Design



Optimization of
Structure



Programming for
Civil Engineering

Summary

Engineering Softwares



Programming Language

- Java
- C++
- Visual Basic
- Python

Languages

- Bahasa Indonesia (Native)
- English (Full Professional Proficiency)
- Javanese (Native/Bilingual)

Certification

 Basic Course Building Modelling With Revit
2025

 Basic Course Building Schedule & Cost
Navisworks
2025

 ITP TOEFL (593/677)
2023

 Ujian Kemahiran Berbahasa Indonesia
(711/1000)
2023

 Microsoft Technology Associate (MTA) :
Introduction to Programming Using Java
2022

 Microsoft Excel (Microsoft 365 Apps and
Office 2019)
2022

 Microsoft PowerPoint (Microsoft 365 Apps
and Office 2019)
2021

Other skill

- Videography and Video editing
- Motion Graphics
- Music Instrument : Drums and piano

Media Related Softwares



Portfolio Table of Contents

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Civil Engineering Portfolio

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Design and Build Prabajaya Tied Arch Bridge Model KJI 2025



42,98 kg
Self Weight



2,494 mm
Deflection Planned



Rp4.432.100,00
Cost



40 minutes
Construction Time

Technical Specification :

Structural Type = Tied Arch Bridge

Length = 3000 mm

Height = 750 mm

Width = 500 mm

Prabajaya Bridge is a Tied Arch Bridge designed to fulfill **optimum, strong, and sustainable aspects**. Prabajaya Bridge is designed for Kompetisi Jembatan Indonesia (KJI), **Indonesia's most prestigious competition for Civil Engineering student**.



Strong

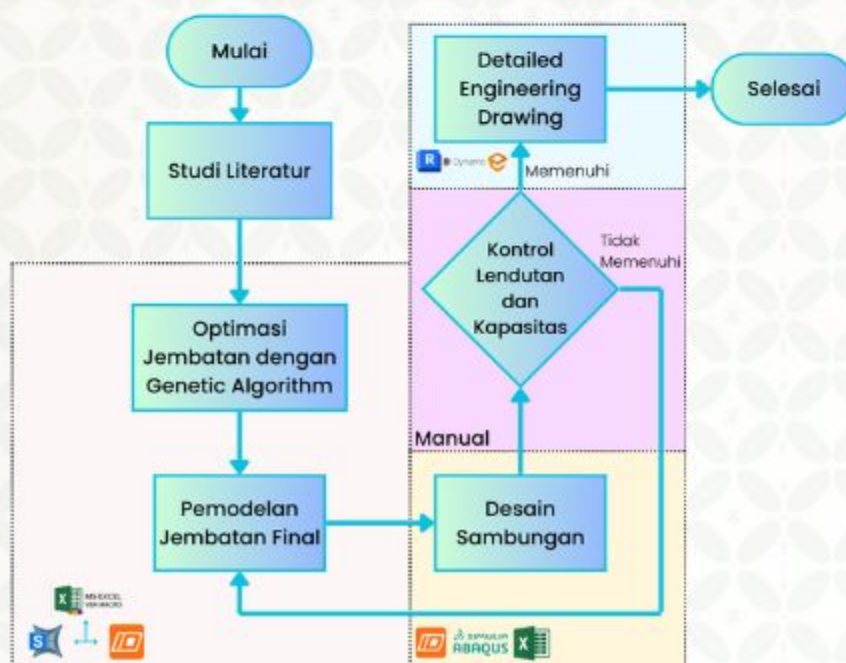


Optimum



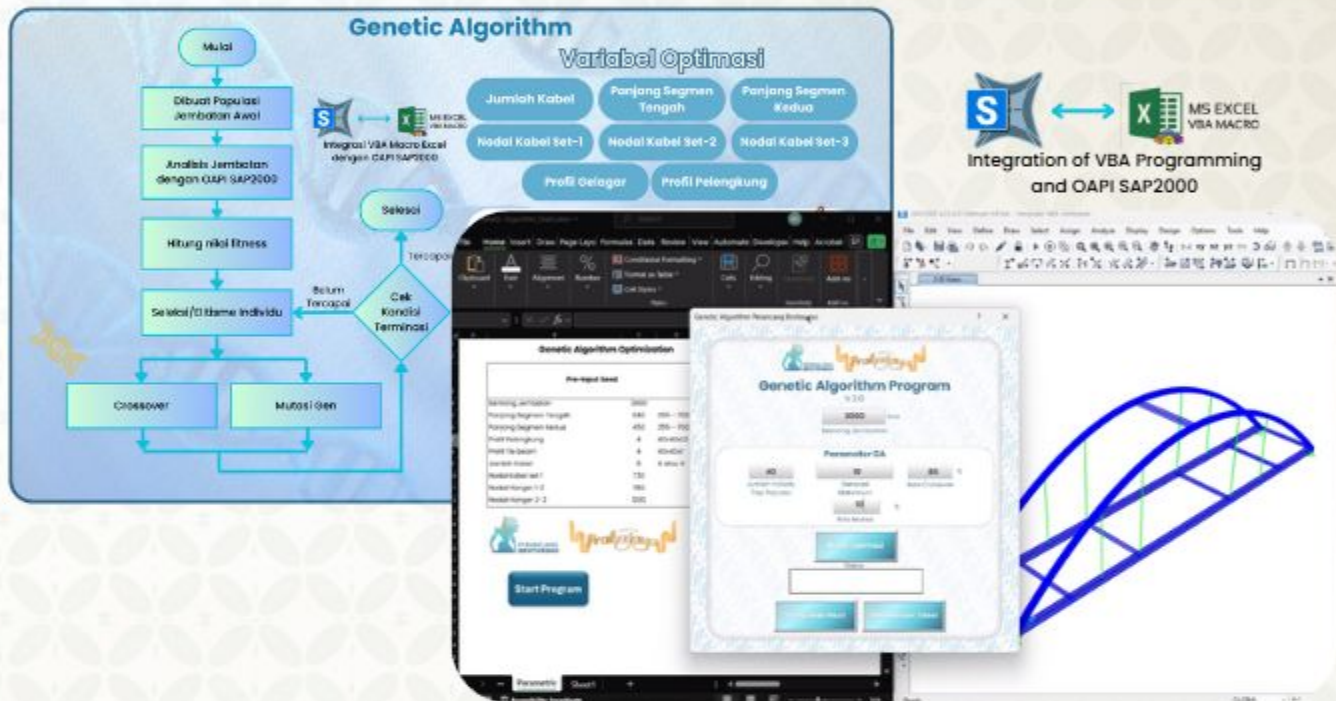
Sustainable

The design methodology incorporates the use of **Building Information Modelling** principles with **integrated softwares** to ensure an effective and efficient design. The design methodology is as follows :



Design and Build Prabajaya Tied Arch Bridge Model KJI 2025

The optimization within the preliminary design incorporates metaheuristic optimization with **Genetic Algorithm (GA)** programming using Visual Basic on OAPI SAP2000

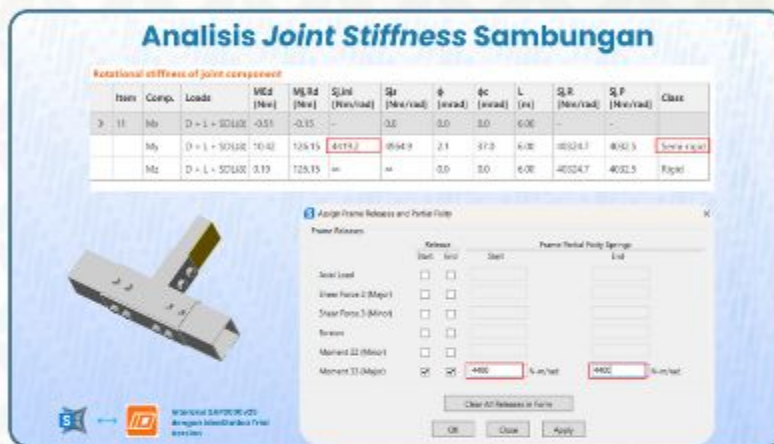


The optimization process analyzed 400 different structural configurations with converged fitness value in the 10th generation of the bridge.

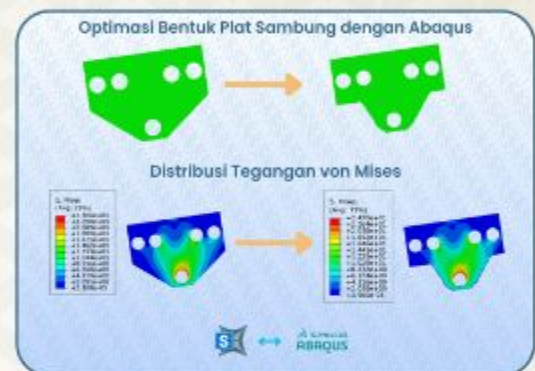


Design and Build Prabajaya Tied Arch Bridge Model KJI 2025

The design also incorporates advanced FEM analysis using IDEASiCa and ABAQUS CAE for **connection design, stiffness analysis, and volume optimization** for accurate analysis and optimum design.



Stiffness Analysis on IDEASiCa and Partial Fixity Release on SAP2000



Connection volume optimization using Topology Optimization on ABAQUS CAE



Bridge Fabrication



Bridge Erection Process



Prabajaya Bridge on site plan

Prabaja Bridge has a deflection of 3.9 mm with a 250 kg load tested on KJI Load site that brings Perancang Brotoseno team secured 2nd place of the competition

The team:

Haikal

- Structural Specialist
- FEM Engineer

Levy

- Bridge Modeller Specialist
- BIM Engineer



Tamiang Agam Network Arch Bridge Design (2025)



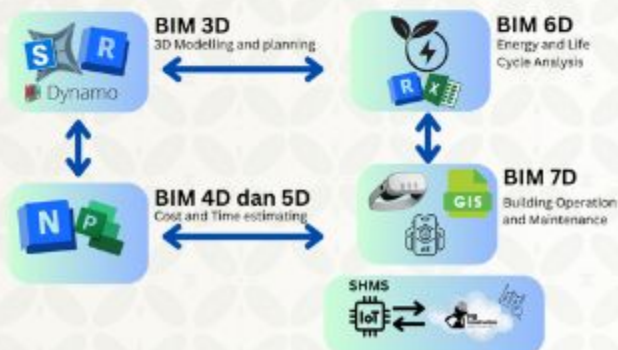
	Bridge Weight
kg	1387.89 ton
	Deflection
	143.5 mm
	Cost
	Rp42.844.037.454,66
	Construction Time
	178 Hari
	Carbon Emission
	868.817 ton

Tamiang Agam Bridge is a bridge designed to be built at Biruen Aceh with 160 m span. Tamiang Agam bridge is a tiebow bridge with network cable configuration designed for Civil Festival competition 2025.

Tamiang Agam is designed with Building Information Modelling and integrated softwares across its project cycle, ensuring its sustainability. The design methodology is as follows :



The use of BIM across its lifecycle is as follows :



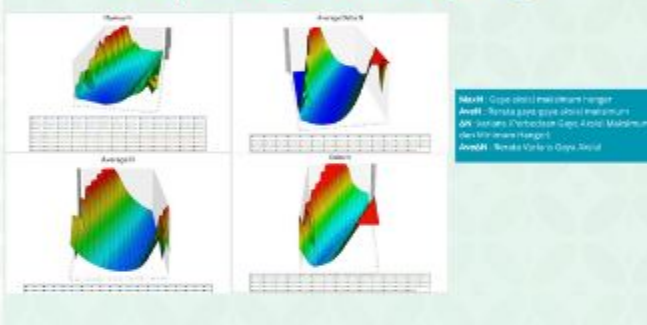
The team:



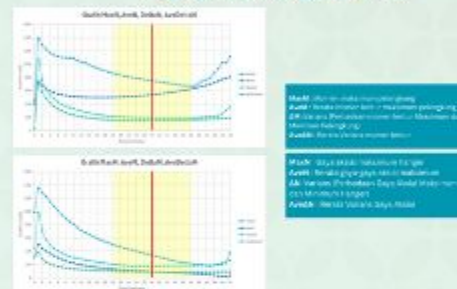
Tamiang Agam Network Arch Bridge Design (2025)

The bridge structure is optimized using the integration of Visual Basic for Application Programming and OAPI SAP2000. The internal forces are minimized but the self-weight is kept reasonably. The optimization result is as follows :

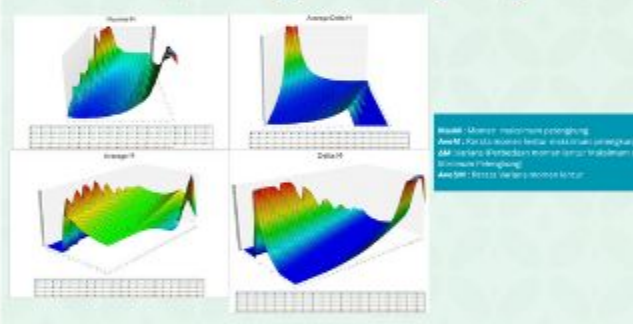
Hasil Optimasi Konfigurasi Constant Angle Change



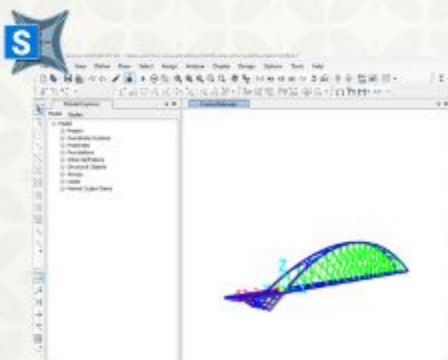
Hasil Optimasi Konfigurasi Radial



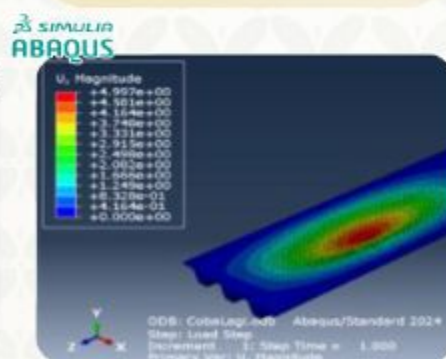
Hasil Optimasi Konfigurasi Constant Angle Change



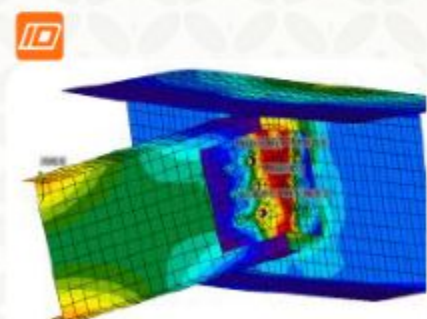
Design Documentation



Global Bridge Design
in SAP2000



Orthotropic Steel Deck
Sub Model Design in
ABAQUS CAE



Steel Connection
Design Using
IDEASTatica

Water Tank Tower Design for Community Service (2025)



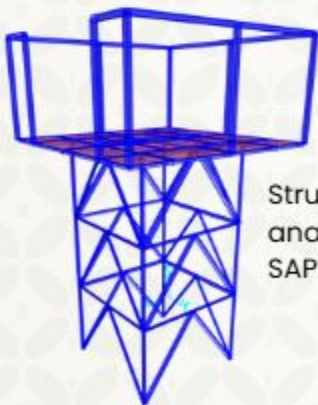
Water tank tower on site

To address the clean water shortage in Cangkring Village, Bantul, the 'Survey dan Perancangan' (Super) community service program has initiated the construction of a well and a water tank to meet the village's needs



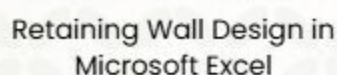
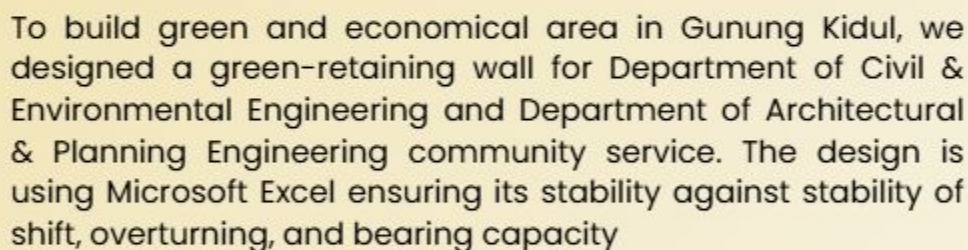
Water tank tower fabrication

The water tank is designed with a steel truss structure, with a height of 3 m above ground. The design process is done by using SAP2000 and Microsoft Excel



Structural design and analysis using SAP2000

Manual Calculation using Microsoft Excel

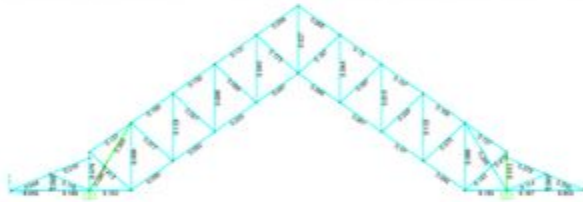


Roof Structure Assessment of UGM Buildings for Solar Panel Installation (2025)



Solar panel installation survey at SGLC FT UGM

The project goal is to ensure the roof structures of all UGM building capacity is still safe for the solar panel installation as a part of UGM Green Energy Program



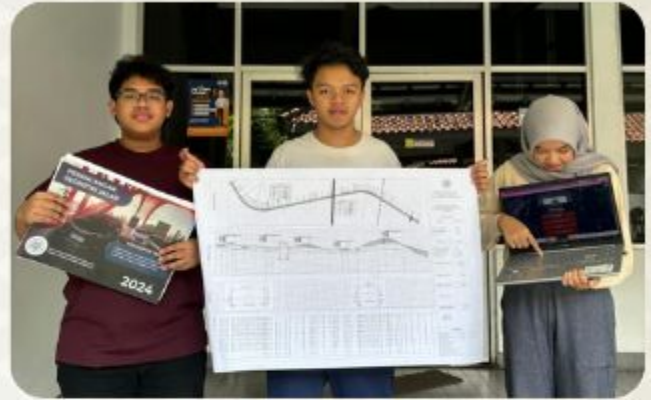
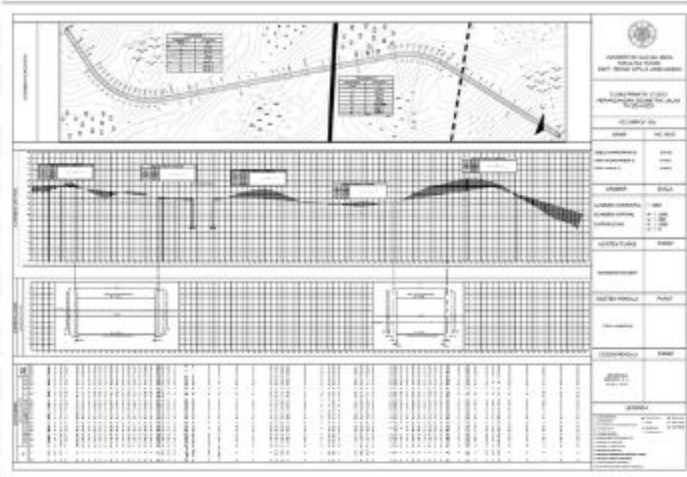
Roof Truss analysis on SAP2000

Nama Bangunan Jenis Rop		Dedung TLC FT UGM Desi Bero	
Analisa Beban			
Analisa Beban Aktif Solar Panel			
Uraian Beban	Data	Satuan	Keterangan
Berat Solar Panel	$W = 21$	kg	Dari Spesifikasi
Spasi konfigurasi	$E = 20$	-	Dari survey
Berat Teropong	$W = 22.35$	kg	
Lebar	$p = 1754$	mm	Dari Spesifikasi
Lebar	$l = 1996$	mm	Dari Spesifikasi
Luas PV	$A_{pv} = p \cdot l = 1.622364$	m^2	
Berat per satuan luas	$q_{pv} = W/A_{pv} = 9.114$	kN/m^2	
Analisa Beban Aktif Dudukan Solar Panel			
Uraian Beban	Data	Satuan	Keterangan
Dudukan Beban			
Berat jenis Beton	$\rho_c = 2300$	kg/m^3	Dari Laporan Struktur
Tebal	$t = 200$	mm	Dari hasil survey
Lebar	$b = 1000$	mm	Dari hasil survey
Lebar	$l = 1000$	mm	Dari hasil survey
$W_{slab} = 47.65 \cdot 10^3 \text{ N}$			

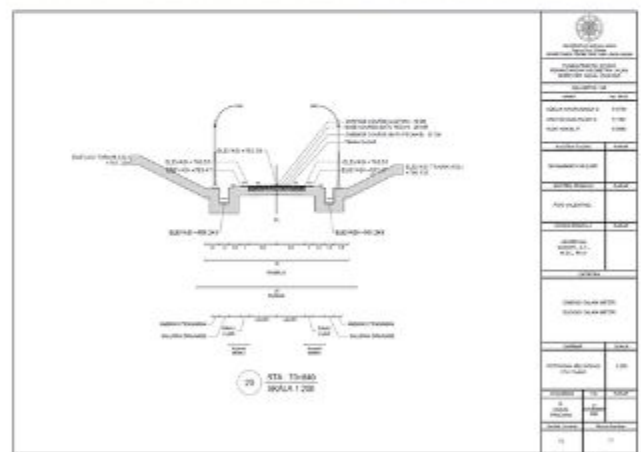
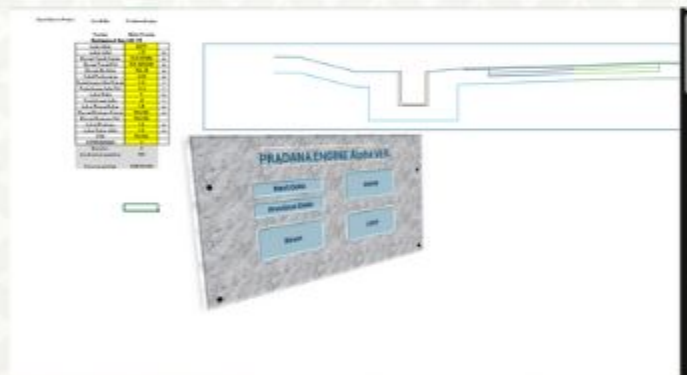
Roof Deck Analysis on Microsoft Excel

The structural analysis is done using SAP2000 and Microsoft Excel by re-modelling roof structure using as built drawings.

Road Geometry Design (2024)



The project was developed as part of the road geometry studio. The design includes: Multicriteria analysis, horizontal and vertical alignment, and drainage system



This project also includes the **creation of cross-sectional drawing automation, plot and stationing automation** in Autodesk AutoCAD using **Visual Basic for Applications** that makes the drawing process faster and more accurate.

Bailey Bridge Deck Handling Recommendation (2024)



Several Bailey Bridges in Sleman were found to be broken, especially the deck structure, due to their age. Lembaga Kajian Fakultas Teknik (LKFT) working collaboratively with PUPR Sleman, gives recommendation for renovation based on survey results and structural calculations.



Laporan Kunjungan Lapangan dan Rekomendasi
Penangan pada Jembatan Padasan, Ngentak dan
Ciyen, di Kabupaten Sleman



Tim Ahli			
No.	Posisi	Nama	Pendidikan
1	Ketua	Dr. Juhar Kusuma S.T., M.T., Ph.D.	S3
2	Anggota Kaya	Dr. M. Suciati, S.T., M.Eng., Ph.D., IPJ, APU	S3
3	Anggota	Kenneth Aremah, ST, MT, Ph.D.	S3
4	Anggota	Dr. H. Supriyanto, S.T., M.T., Ph.D.	S3
5	Anggota	Angga L. S. Satrio, S.T., M.Eng., Ph.D.	S3
6	Anggota	Muhammad Fauzi Darmawan, S.T., M.Eng.	S3
7	Asisten	Yusuf Hidayat, S.T., M.Eng.	S2
8	Asisten	Khotuloh, Surya Adhika, S.T.	S1
9	Asisten	Ayulita N. Musthofa	SLTA
10	Asisten	Ermanuelle Lery Hanjanto	SLTA
11	Asisten	Riko Haka Prabono	SLTA

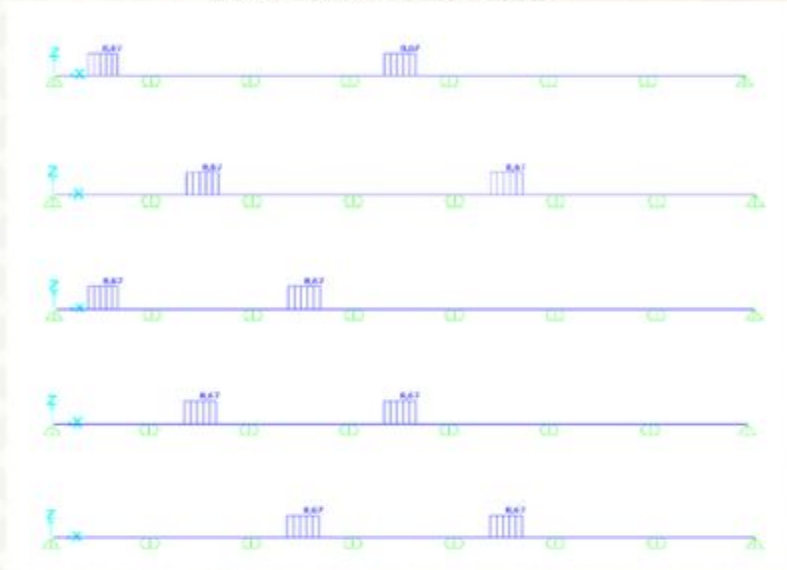


Roles in Project

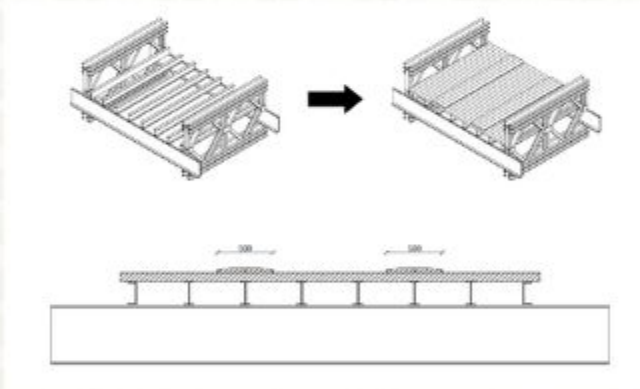
- Conducted survey to calculate the existing dimensions of the structures
- Assess and give recommendation of deck renovation using two options, wood deck and steel deck using technical calculation in SAP2000 and Microsoft Excel.

Bailey Bridge Deck Handling Recommendation (2024)

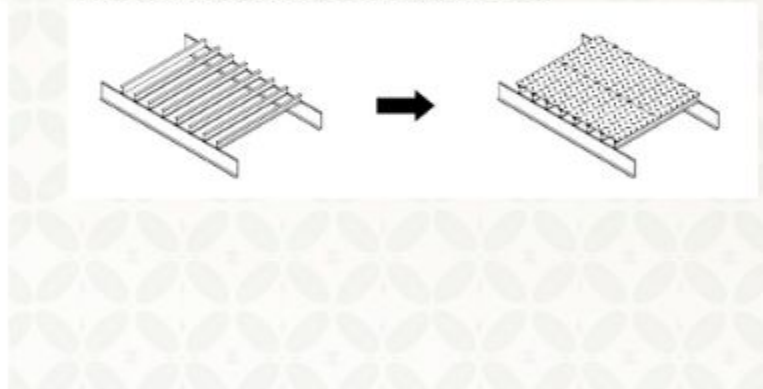
Load Input in SAP2000



First Alternative : Wood Deck



Second Alternative : Steel Deck



Capacity Calculation Using Excel

Tabel 4. Desain Plat Lantai Kayu

Alternatif 1 (Kayu)			
Pelapis	Bitumen 3 mm		
Tipe	Bangkerai E12		
Tinggi (mm)	80		
Lebar (mm)	120		
Lentur	σ_b (Mpa)	F_b' (Mpa)	DCR (σ_b/F_b')
	5.24	11.66	0.45
Geser	τ_v (Mpa)	F_v' (Mpa)	DCR (τ_v/F_v')
	1.13	1.28	0.88

Tabel 7. Desain Pelat Lantai Baja

Alternatif 2 (Diamond Plate)			
Pelapis	Bitumen 3 mm		
Tipe	Diamond Plate Baja Karbon Q235		
Tebal (mm)	6		
Lebar (mm)	1500		
Lentur	σ_u (Mpa)	$\Phi F_y'$ (Mpa)	DCR ($\sigma_u/\Phi f_y'$)
	156.91	211.50	0.74
Geser	V_u (kN)	ΦV_n (kN)	DCR ($V_u/\Phi V_n$)
	15.21	117.42	0.13

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[instagram.com/haikallpradana](https://www.instagram.com/haikallpradana)



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